

Thursday, 1 January 2026

What is Amazon EC2?

Amazon EC2 (Elastic Compute Cloud) is an AWS service that lets you **rent virtual servers in the cloud** to run applications, websites, databases, or any software—without buying physical hardware.

Think of EC2 as **a computer in the cloud** that you can start, stop, resize, and pay for only when you use it.

Why is EC2 used?

EC2 is mainly used to:

- Host **websites & applications**
- Run **backend services**
- Perform **data processing**
- Create **test/dev environments**
- Support **DevOps & CI/CD pipelines**

Key EC2 Concepts (Very Important)

1 Instance

An **EC2 instance** is a virtual server.

Example:

EC2 instance = Linux/Windows machine in AWS

2 Instance Types

Defines **CPU, RAM, storage, and network performance**

Common types:

- **t2 / t3** – Low-cost, general purpose (testing)
- **m5** – Balanced performance
- **c5** – Compute optimized (CPU heavy apps)
- **r5** – Memory optimized
- **g4** – GPU based (ML, AI)

Example:

`t2.micro` → 1 vCPU + 1 GB RAM (Free tier)

3 AMI (Amazon Machine Image)

An **AMI** is a template used to create EC2 instances.

Examples:

- Amazon Linux
- Ubuntu
- Red Hat
- Windows Server

4 Storage Options

- **EBS (Elastic Block Store)** – Persistent disk (like hard drive)
- **Instance Store** – Temporary storage

5 Key Pair

Used for **secure login** to EC2.

- `.pem` file → SSH login
- Required for Linux instances

Example:

```
ssh -i key.pem ec2-user@public-ip
```

6 Security Group

Acts as a **virtual firewall**.

Controls:

- Inbound traffic (SSH, HTTP, HTTPS)
- Outbound traffic

Example:

Allow SSH (22) from your IP
Allow HTTP (80) from anywhere

7 Elastic IP

A **static public IP** for EC2.

Used when you don't want IP to change after restart.

EC2 Pricing Models

You pay **only for what you use** 🕒

1. **On-Demand** – Pay per hour/second
2. **Reserved Instances** – Long-term, cheaper
3. **Spot Instances** – Very cheap, but can stop anytime
4. **Dedicated Hosts** – Physical server for one customer

EC2 Lifecycle

Launch → Start → Stop → Reboot → Terminate

- **Stop**: Data on EBS remains
- **Terminate**: Instance deleted permanently

Simple EC2 Use Case

👉 Hosting a website:

1. Launch EC2 (Ubuntu)
2. Install Apache/Nginx
3. Open port 80 in Security Group
4. Access using public IP

EC2 in DevOps (Interview Point)

- Used with **Auto Scaling**
- Integrated with **Load Balancer**
- Part of **CI/CD pipelines**
- Supports **Infrastructure as Code** (Terraform, CloudFormation)

In One Line (Interview Ready)

Amazon EC2 is a scalable, on-demand virtual server service in AWS that allows users to run applications in the cloud with full control over computing resources.